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The Iris Dataset: An accidental Case Study in pedagogy, methodology and real-world implications

*** Author's Note**

The consultant's name has been redacted for privacy and professional courtesy. This case study examines two parallel analytical workflows as expressions of distinct STEM training lineages, not the individuals who performed them. Any consultant-generated artifacts appear only in anonymized form to illustrate methodological contrasts.

"One class is linearly separable from the other 2; the latter are not linearly separable from each other."⁵

Introduction

The Iris dataset is ubiquitous in introductory statistics and machine learning. It is typically presented as a parable of clean categorical separation: three distinct species (*Iris setosa*, *I. versicolor*, and *I. virginica*) perfectly distinguishable by their measurements. However, a closer examination reveals a more complicated truth. While *I. setosa* is distinct, *I. versicolor* and *I. virginica* overlap significantly—a nuance explicitly noted in the dataset’s canonical documentation yet frequently erased in introductory pedagogy. The Iris dataset’s documentation explicitly states that two species overlap (“One class is linearly separable from the other; the latter are not linearly separable from each other”), yet it serves as the canonical example of perfect classification in statistics and machine learning education.² This contradiction is not accidental—it reveals how statistical pedagogy systematically erases empirical complexity to preserve the myth of clean categorical boundaries.

Science is methodology, but methodology is not neutral. The tools we use to observe and interpret the world are shaped by the values and assumptions of their creators. Statistics, in particular, emerged from efforts by a small cohort of white men (non exhaustive: Galton, Pearson, and Fisher) seeking to validate the differences between white and black their ideologies insisted upon. Creating a measure would shift opinion into the domain of objectivity. Creating measures, scales and standards required a “from” or a point to measure proximity. How else would one know how close one was or was not to the mark? An idealized white, cisgender male body to quantify human “difference” was the benchmark. Everybody nonwhite, non-cisgender male has been measure to this. Medicine, even today still has its benchmarks (within normal limits) as a white cis male body. Oxygen saturation measuring, kidney function and pain presentation are not remnants of some past misstep.

What often goes unexamined is how this logic of deviation from the presumed standard (central tendency) functions as alpha and omega by mirroring and reinforcing systems of social classification and hierarchy. Although Fisher later distanced himself from some extreme forms of biological determinism, the statistical frameworks he helped formalize were nonetheless forged in an era when eugenics, typological thinking, and colonial science were dominant paradigms ¹.

The core research question began simply: *Are there significant differences between the species, and can they be accurately predicted?* However, the divergence between one analysts' findings conducted through a standard predictive lens (perfect separation) and a second analysis (structural overlap) reframed the inquiry.

Because the consultant and I worked independently, our workflows accidentally became a natural experiment in statistical pedagogy. One analysis followed the predictive pipeline toward separability; the other used an exploratory-diagnostic lens to surface structural overlap. Their divergence was not a disagreement but the case study itself: a demonstration of how training shapes what analysts see and what they discard.

Through a critical re-analysis, this report demonstrates how the choice of lens determines the visibility of the data. When we prioritize algorithmic separation over diagnostic complexity, we risk creating models that are mathematically accurate but structurally false—a legacy that extends from Fisher's methodology to modern debates on biological determinism in population health, policy and our lives.

A Note on Positionality and Reflexive Methodology

I approach this dataset through a specific critical lens: one that seeks to uncover how ideological commitments become embedded in seemingly neutral statistical practice. This lens necessarily guides my methodological choices. Unlike the uncritical application of statistical procedures, I explicitly surface my theoretical commitments, following Kincheloe's (2008) notion of "critical transparency." I view the Iris dataset not as a neutral "Hello World" for machine learning, but as a seed crystal of a particular statistical ideology—one that prioritizes separation over overlap, purity over mixture, and prediction over understanding.

I.I. Methods

2.1 Data Source The Iris dataset, sources on the UC Irvine Machine Learning homepage. The dataset contains 150 observations of iris flowers, divided equally among three classes/groups: *Setosa*, *Versicolor*, and *Virginica*. Four features were measured for each sample: Sepal Length, Sepal Width, Petal Length, and Petal Width ⁵.

2.2 Exploratory Data Analysis (Client) To establish a baseline, an Exploratory Data Analysis (EDA) prioritizing discovery and visualization was conducted ([EDA here](#)). This included generating summary statistics, pairwise scatterplots to identify multicollinearity, and unsupervised K-Means clustering to test natural grouping without labels (see [appendix](#) for tables and secondary figures to reduce clutter in the main document)

Figure 1 Distribution of Floral Measurements by Species. The horizontal orientation reveals a critical pattern: consistent whisker lengths across species suggest that within-group variation is similar, differing primarily in central tendency rather than distribution shape. This visual impression of "shifted" but similarly shaped distributions—rather than fundamentally different types—challenges the strict assumptions of normality and independence required for the consultant's standard ANOVA approach. Note specifically the substantial overlap in Sepal Width, which contradicts the hypothesis of clean categorical separation.

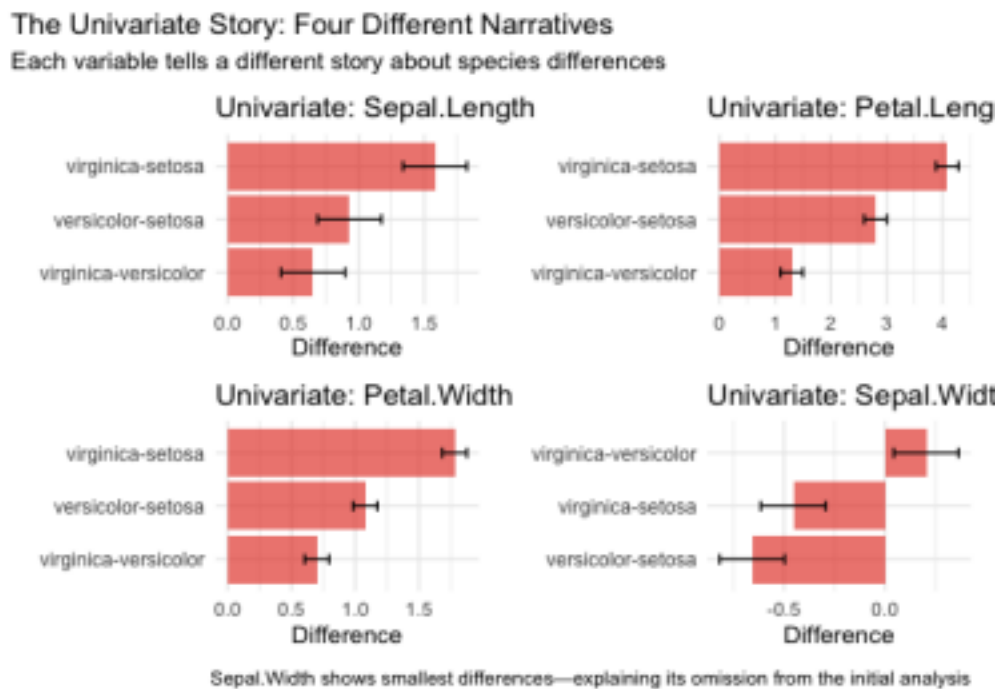
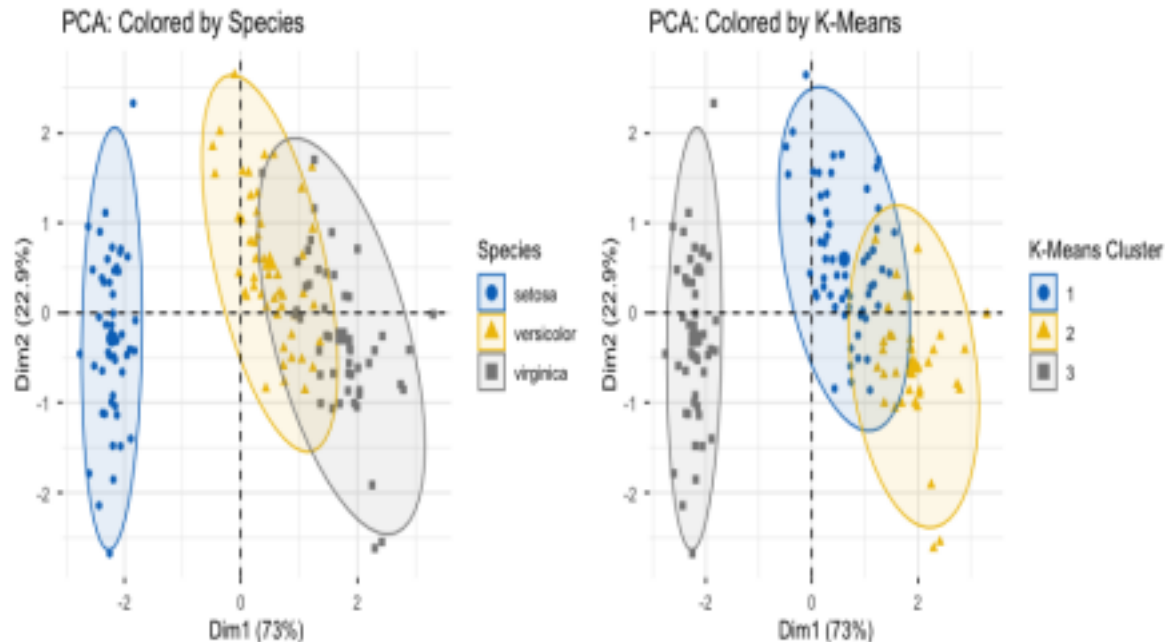


Figure 2. Principal Component Analysis (PCA) Colored by Species and K-Means Cluster Assignment
Left panel: PCA of the Iris dataset colored by species (Setosa, Versicolor, Virginica), showing clear separation, particularly along PC1. Right panel: The same PCA plot colored by K-means cluster assignment (k=3), demonstrating approximate correspondence between unsupervised clusters and true species groups. Ellipses indicate 95%



confidence intervals for each group ⁴.

2.3 Inferential Analysis (Consultant) The consultant operated independently drawing from a meeting with the client and left with background information, EDA, research question, and suggested methodologies for consideration in their approach. The consultants selected methodology involved ([Output](#)):

Linear Regression: Three separate linear models were fitted for Sepal Length, Petal Length, and Petal Width. Notably, Sepal Width was omitted from this testing phase.

Principal Component Analysis (PCA): A duplicative PCA was conducted, followed by a linear regression of the first principal component against species.

Multinomial Logistic Regression: A supervised classification model was fit to predict species based on floral measurements.

2.4 Remedial Inferential Analysis (Client) Upon receiving the consultant's report, the client identified significant methodological gaps and performed the following analyses([Here](#)):

- **MANOVA:** To account for the relationship between dependent variables, replacing the isolated linear regressions.

- **Tukey's HSD:** To provide the necessary post-hoc pairwise comparisons that the consultant omitted.
- **Linear Discriminant Analysis (LDA):** To model the separation of classes using a method appropriate for the data structure.

III. Results

The investigation revealed a stark contrast between the illusion of clean separation presented in the consultant's initial report and the nuanced reality uncovered through reanalysis.

3.1 The consultant's analysis painted a picture of definitive separability, but this conclusion relied on systemic methodological choices:

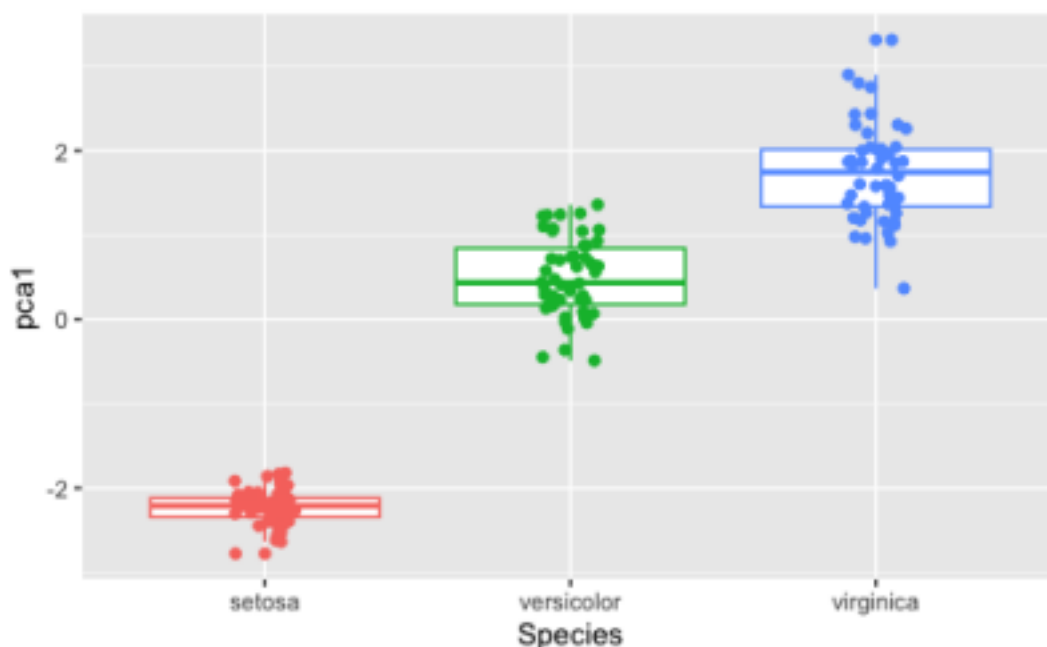
The "Significance" of Omission: The consultant ran linear regressions (functionally equivalent to ANOVA) on three of the four variables: Sepal Length, Petal Length, and Petal Width. All returned high significance ($p < 2e-16$). However, the consultant excluded Sepal Width—the one variable that shows significant overlap between *Versicolor* and *Virginica*. By removing the "messy" data point, the analysis artificially inflated the clarity of the species boundaries.

The Conflation of Prediction and Inference: While linear regression and ANOVA are mathematically equivalent (both represent the General Linear Model), the consultant utilized regression purely as a classification tool rather than a hypothesis test. A rigorous ANOVA framework asks, "*Do specific group means differ?*" and mandates assumption checks and post-hoc analysis. In contrast, the consultant's approach implicitly asked, "*Can I separate these groups?*" By prioritizing separation over inference, the analysis treated the biological overlap in *Sepal Width* not as a diagnostic finding, but as "noise" to be discarded in pursuit of a cleaner, albeit less accurate, model.

Circular PCA Logic: The consultant performed a regression using the first Principal Component (PC1) as the dependent variable against Species. Since PC1 is mathematically constructed to maximize variance (which is driven by species differences), this test was circular—it could only show that the species were different because the math defined them

as such³.

Figure 1 (Consultant Output). Principal Component Analysis (PC1) scores by Species. Principal Component Analysis (PCA) and visualized the scores of the first component (PC1) by species. PC1 accounts for 72.96% of the variance and effectively maximizes the separation between groups. This visualization serves as a summary of the primary variation, distinct from the bi-dimensional structural overlap (PC1 vs PC2) identified in the initial Exploratory Data Analysis ([see Appendix](#))



The Broken Classifier: The consultant presented a multinomial logistic regression with a "perfect" confusion matrix for the training set. However, a review of the raw code revealed that the model threw multiple warnings: *fitted probabilities numerically 0 or 1 occurred* and the algorithm *did not converge*. These warnings indicate "complete separation"—the model failed to converge because *Setosa* is too distinct, while the other classes overlapped. Rather than addressing this breakdown, the warnings were suppressed, and the "results" were presented as valid.

Post-hoc tests would have revealed that versicolor/virginica do not significantly differ on sepal width—a direct contradiction to the 'clean separation' narrative. The system requires analysts to stop at $p < 0.001$ because deeper interrogation exposes the dataset's explicit warning about overlap. This isn't incompetence; it's the logical endpoint of teaching stats as 'find significance, then stop.' In population health, this produces race-based models that

'predict' disparities while ignoring structural drivers—because asking 'which groups differ?' would reveal continuous health gradients shaped by systemic racism. We can just as easily place gender in this continuum as well.

In biology, we know *versicolor* and *virginica* are overlapping taxa—Fisher's dataset was curated to hide this. His logistic regression 'succeeds' (100% accuracy) because it treats *Setosa*'s separability as the goal, not a diagnostic red flag. But in population health, if we treated race categories like Iris species—forcing separation while ignoring overlap—we produce models that 'accurately' predict Black infant mortality via race, while obscuring how structural racism shapes continuous health gradients (a preference for aggregation over disaggregation causes erasure of populations).

3.2 Client Re-Analysis

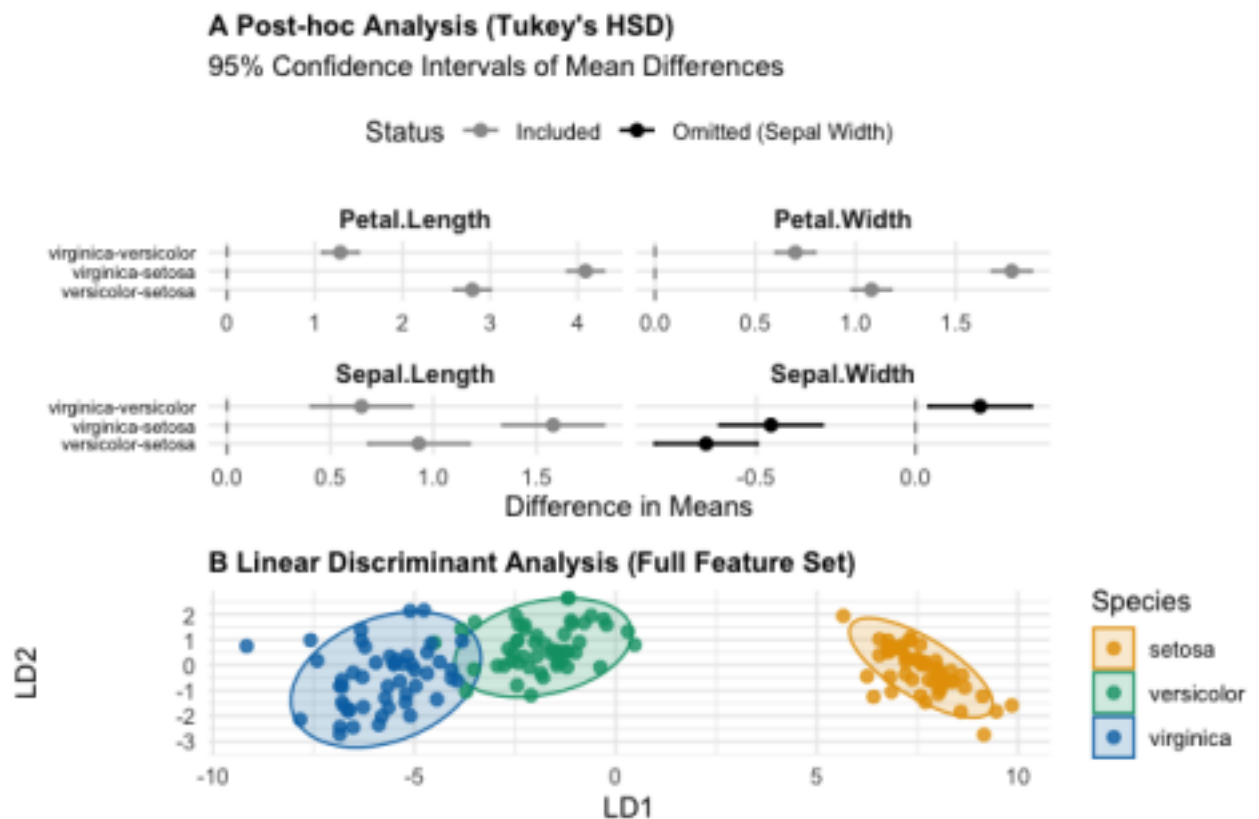
Client conducted omitted inferential tests

- **MANOVA and Tukey HSD:** Unlike the isolated regressions, the MANOVA shows significant differences overall ($p < 0.001$). Crucially, the subsequent Tukey HSD tests revealed that while *Setosa* is distinct across all metrics, the difference between *Versicolor* and *Virginica* is nuanced, particularly in Sepal Width, where the mean difference is statistically smaller and distributions overlap.

To visualize the structure of the data without artificial variable selection, the client performed a Linear Discriminant Analysis (LDA) using the complete four-variable feature set. The LDA provides a dimensionality reduction optimized for class discrimination while retaining the biological context of all measured traits

Figure 3. Reintegration of excluded features reveals significant biological structure

deployed in remedial analysis. (A) Tukey's HSD pairwise comparisons of mean differences for all four morphological traits. Bars represent 95% confidence intervals. Intervals that do not intersect zero indicate statistical significance ($p < 0.05$). Variables colored blue were retained in the consultant's model; *Sepal Width* (red) was omitted. Note that despite its exclusion, *Sepal Width* shows significant differences for multiple species pairs. **(B)** Linear Discriminant Analysis (LDA) projection utilizing the full 4-variable feature set. Ellipses indicate 95% confidence regions for *I. setosa* (red), *I. versicolor* (green), and *I. virginica* (blue). The latent structure confirms distinct clustering of *I. setosa* along LD1, while honestly depicting the biological proximity and slight overlap between *I. versicolor* and *I. virginica*, a reality obscured by aggressive feature selection



- **Linear Discriminant Analysis (LDA):** Examination of the Linear Discriminant coefficients reveal the magnitude of the exclusion error. On the first discriminant axis (LD1), which accounts for 99.1% of the between-group variance, *Sepal Width* exhibits a coefficient of **1.53**, nearly double that of the retained *Sepal Length* (**0.83**).
- While *Petal* dimensions drive the negative direction of the axis (-2.20, -2.81), *Sepal Width* is the primary feature defining the positive direction. By excluding it, the consultant's model did not merely simplify the data; it removed the strongest positive discriminator in the feature set, forcing the model to rely on the weaker signal of *Sepal Length*.

IV. Conclusion: Methodology as Ideology

In population health, this produces race-based models that "predict" disparities while ignoring structural drivers. By asking "*can I separate groups?*" (classification) rather than "*why do these groups differ?*" (inference), analysts encourage the use of race as a biological proxy.

We see this logic fail in high-stakes contexts, such as using race to predict infant mortality while ignoring social determinants, and in everyday policy, such as the debate in Wisconsin over when elders should stop shoveling snow. This is not merely a statistical shortcut; it is the very bedrock of biological determinism/essentialism.

By treating Age as a binary cutoff for "shoveling ability," we ignore functional health. By treating race as a binary cutoff for health risk, we ignore racism. In both cases, the model creates a dangerous fiction: it reifies the category (age/race) as the cause, effectively hiding the actual mechanism (health/racism). This protects the system from scrutiny while pathologizing the individuals within it.

It is critical to acknowledge that the consultant's approach was not technically 'incorrect' within the paradigm of standard data science training; rather, it was optimized for a different objective. The consultant's methodology prioritized parsimony and separation—the hallmarks of efficient predictive modeling. By this metric, the removal of 'noisy' variables like *Sepal Width* is a feature, not a bug, as it creates a cleaner, more definitive classification boundary⁹.

However, the client's objective as an equity-centered scientist was not classification, but diagnosis and structural understanding. Through this lens, the 'noise' of overlapping groups is not waste to be discarded, but the primary signal of biological (or social) reality. The friction in this collaboration did not arise from errors, but from the distinct lineages of STEM training we came from: one shaped by the predictive logic of statistics, the other shaped by the structural and contextual demands of population health. Both approaches

were internally coherent; their divergence simply revealed how training directs attention toward different forms of truth. The consultant succeeded in building a map that was easy to read; the client required a map that was true to the terrain¹⁰. This case study demonstrates that the Iris dataset is not a neutral teaching tool; it is a seed crystal for a statistical ideology that prioritizes separation over structural reality. The client and the consultant became a textbook example of this legacy in 2025, a convergence so precise it could not have been improved by orchestration. It's very lack of orchestration is what makes it so revealing. Moving forward requires training equity-centered analysts not just to run code, but to recognize when a proxy is masking a mechanism¹¹. We must learn to retain the "Sepal Widths"—the messy, overlapping variables—because that is often where the structural truth resides.